Application of Graphene-based 2D Materials in Energy Storage Devices

presentation submitted by:

Name : **Aditya Vishwakarma**

University Roll No. : **002530701010**

Subject : **Seminar**

Semester : **1st**

Academic Year : **2025-2026**

Course : **MTech in Nanoscience and Technology**

Department : **School of Material Science and Nanotechnology**

Introduction

The modern world cannot function without energy. Energy storage devices are crucial for technological advancement in any domain.

Two most common energy storage devices are:

1. BATTERIES
2. CAPACITORS

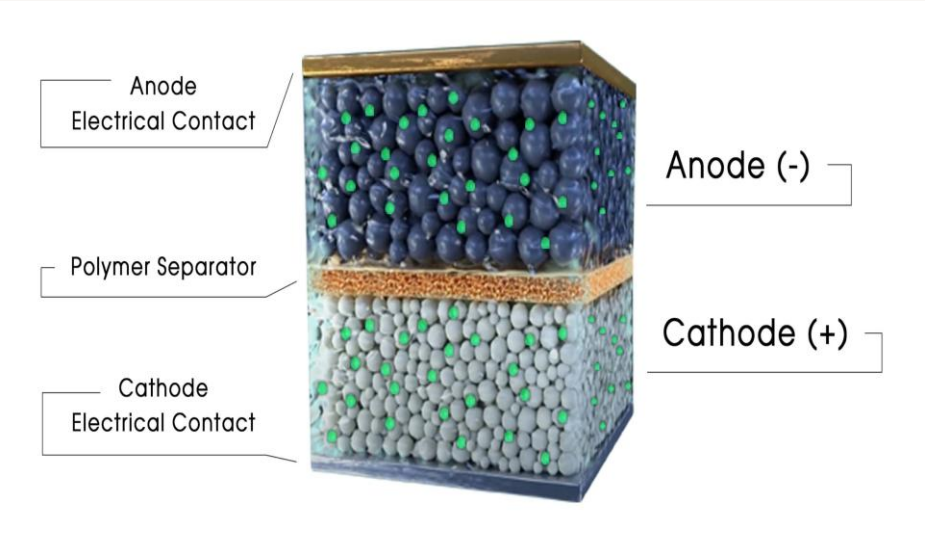
As widespread as these devices may be, they have issues.

Lithium-ion Batteries

1. The battery structure consists of a **cathode** (positive electrode, typically metal oxide), an **anode** (negative electrode, typically graphite), an **electrolyte**, and a **separator**.

2. Li+ ions move from the **anode** through the electrolyte and separator to the **cathode**. The corresponding flow of electrons through the external circuit creates the electric current.

3. An **external current** forces Li+ ions to **move back** from the **cathode** to be stored (intercalated) in the **anode** structure.



Their small size, lightweight nature, and high energy density make them ideal for:

- Smartphones and tablets
- Laptops
- Wearable devices (smartwatches, fitness trackers)
- Portable power banks

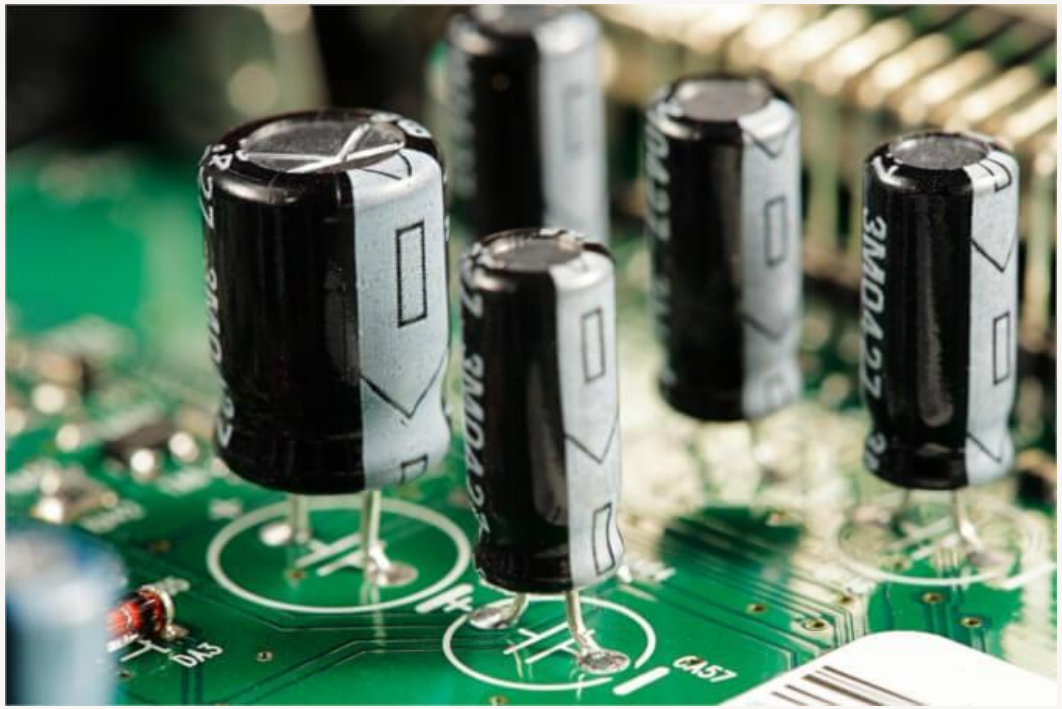
Capacitors

1. It consists of **two conductive plates** (usually metal) separated by a **dielectric** (an insulating material like air, paper, or ceramic).

1. When the capacitor is fully charged, one plate has an excess of electrons (making it the **negative** plate), and the opposite plate has a deficit of electrons (making it the **positive** plate). This separation of charge creates the stored energy.

1. When you connect the two plates with a conductor (like a wire and a load, such as a lightbulb), you create a path for the stored charge to equalize.

1. Capacitors are widely used in **AC-DC Chargers, Computer Motherboards, Touchscreens.**



Issues With Traditional Devices

Safety Concerns

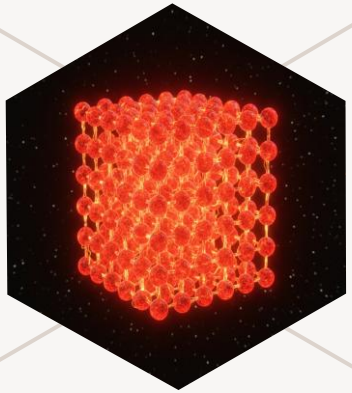
Lithium-ion batteries are susceptible to thermal runaway and dendrite formations.

Limited Lifespan

Lithium-ion batteries swell up when they degrade.

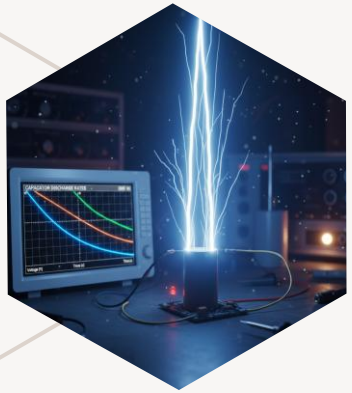
Operational Conditions

Performance of batteries and capacitors is very temperature dependent.



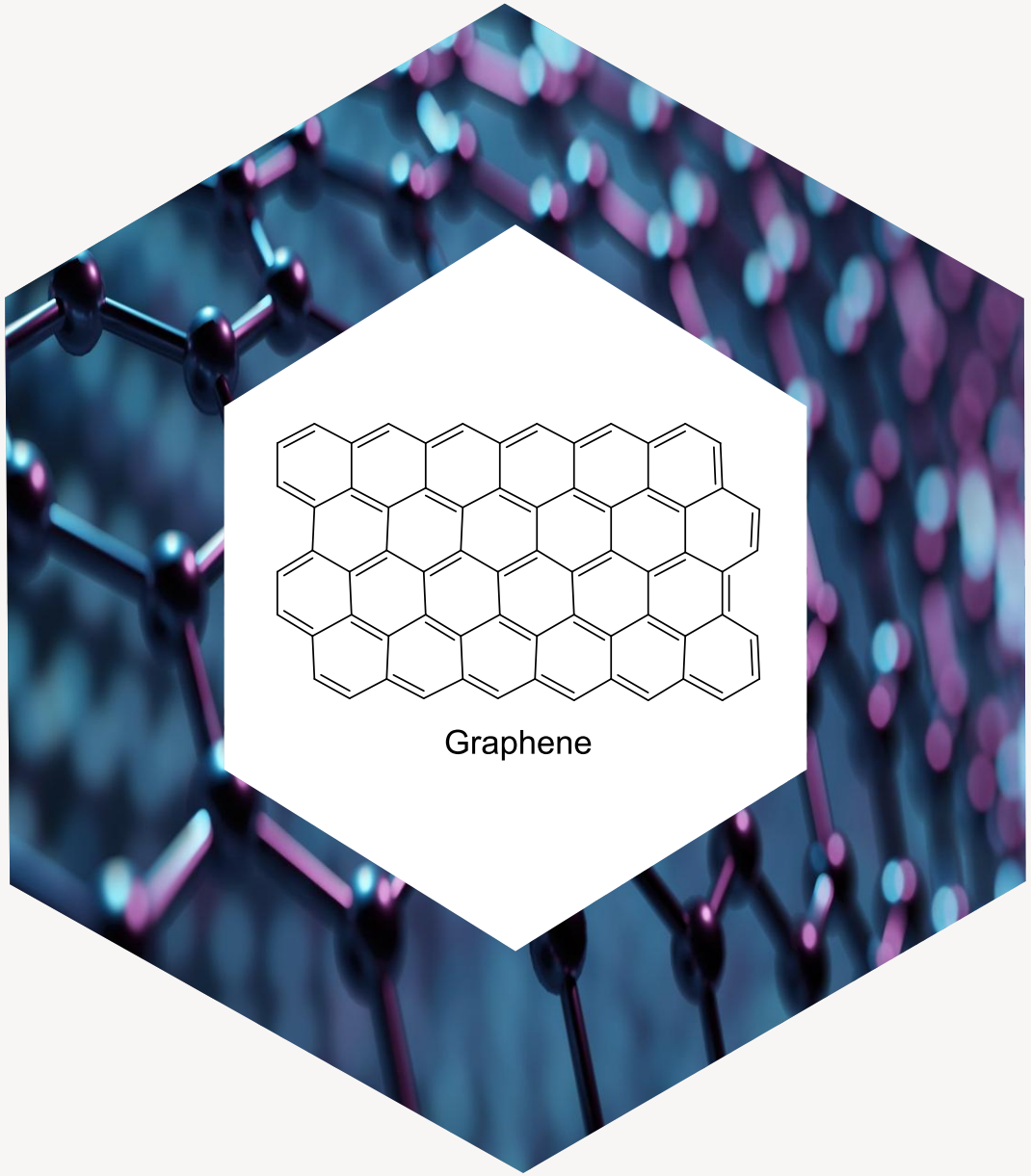
Energy Density

The energy density of batteries and capacitors are less compared to their upgraded counterparts.



Discharge Rates

The discharge rates of capacitors is very fast making them unsuitable for long term use.



What is Graphene?

1. Graphene is a two-dimensional hexagonal carbon atom lattice that is exactly one atom thick.
2. Derived from Graphite which is basically stacked graphene.
3. Extremely lightweight, flexible and transparent.

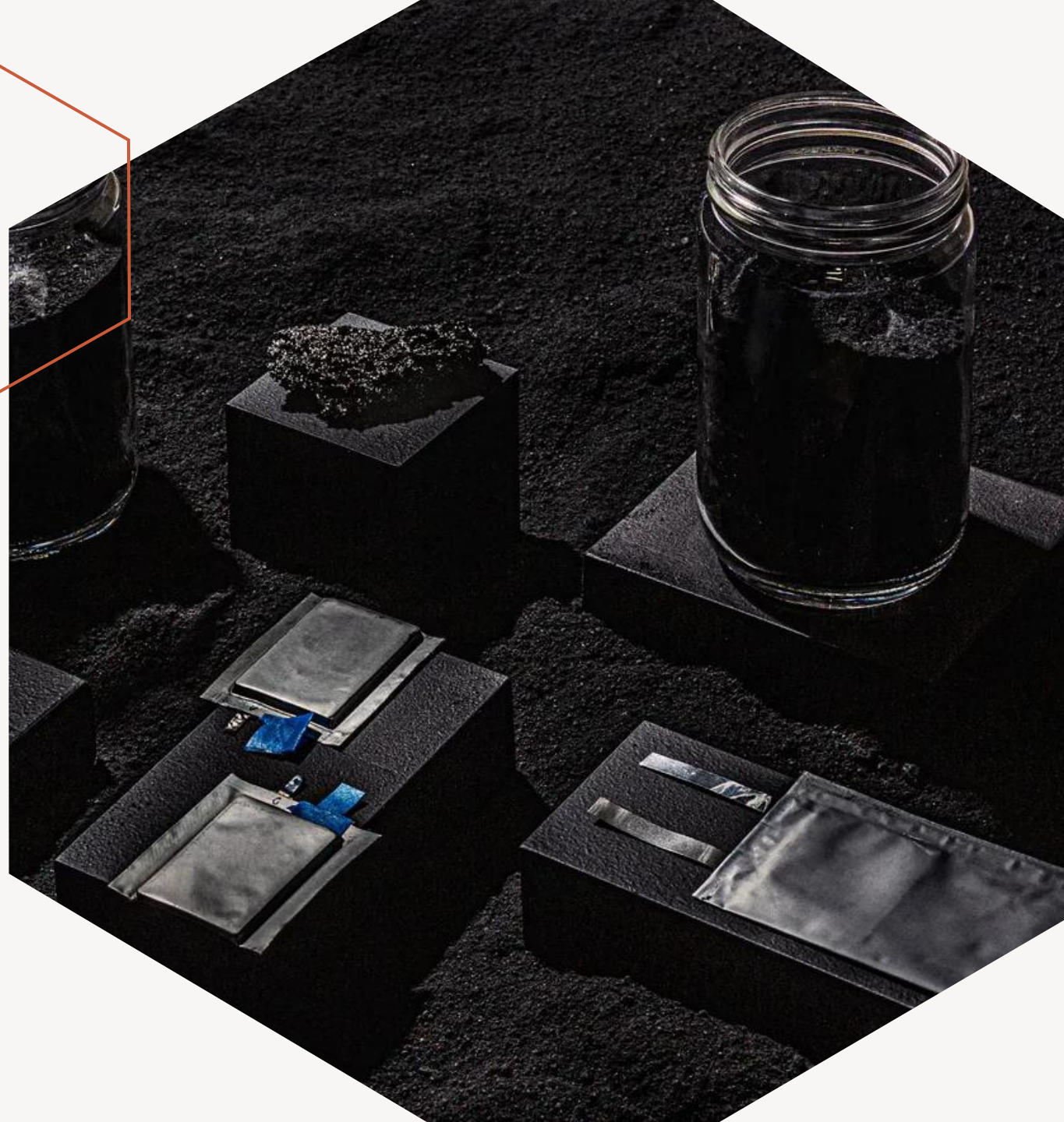
Why Graphene in Energy Storage?

- Faster Charging Times
- More Energy Storage Capabilities
- Highly Durable Devices
- No More Overheating in Devices
- Lighter Devices

Electrode Enhancement

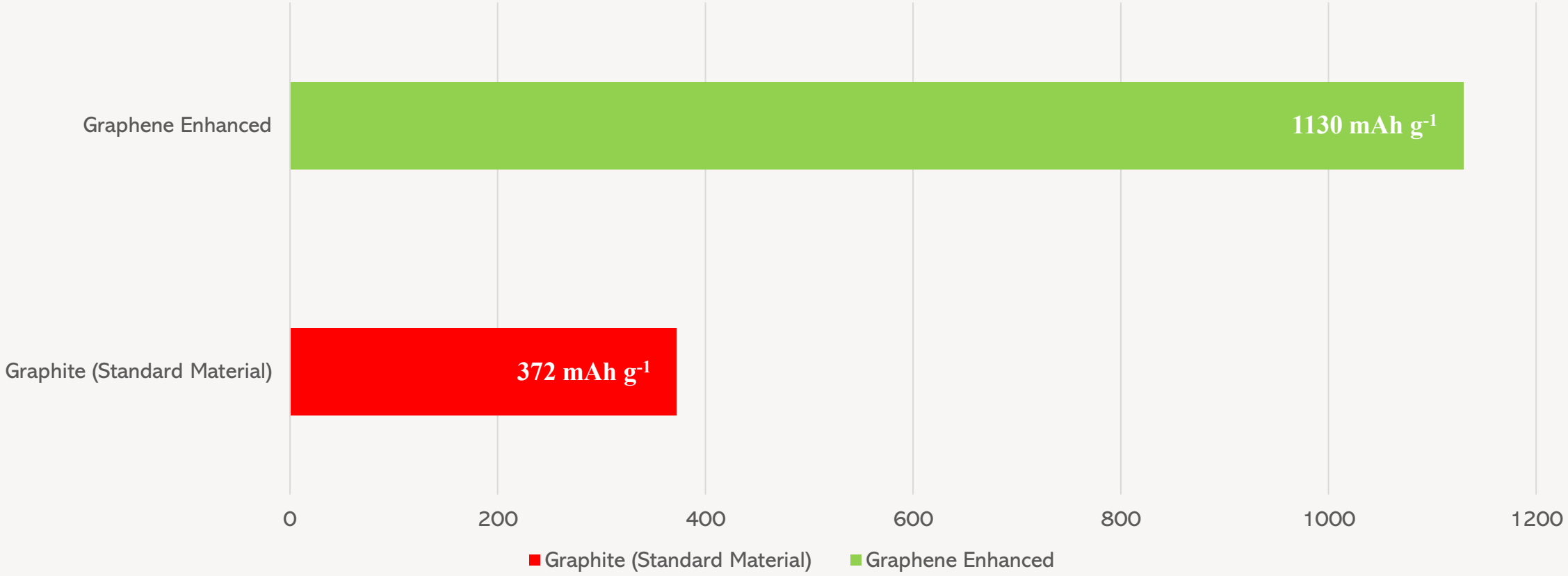
Graphene is added to battery and capacitor electrodes to enhance the electrode’s properties.

1. Electrical Conductivity
2. Durability
3. Stability



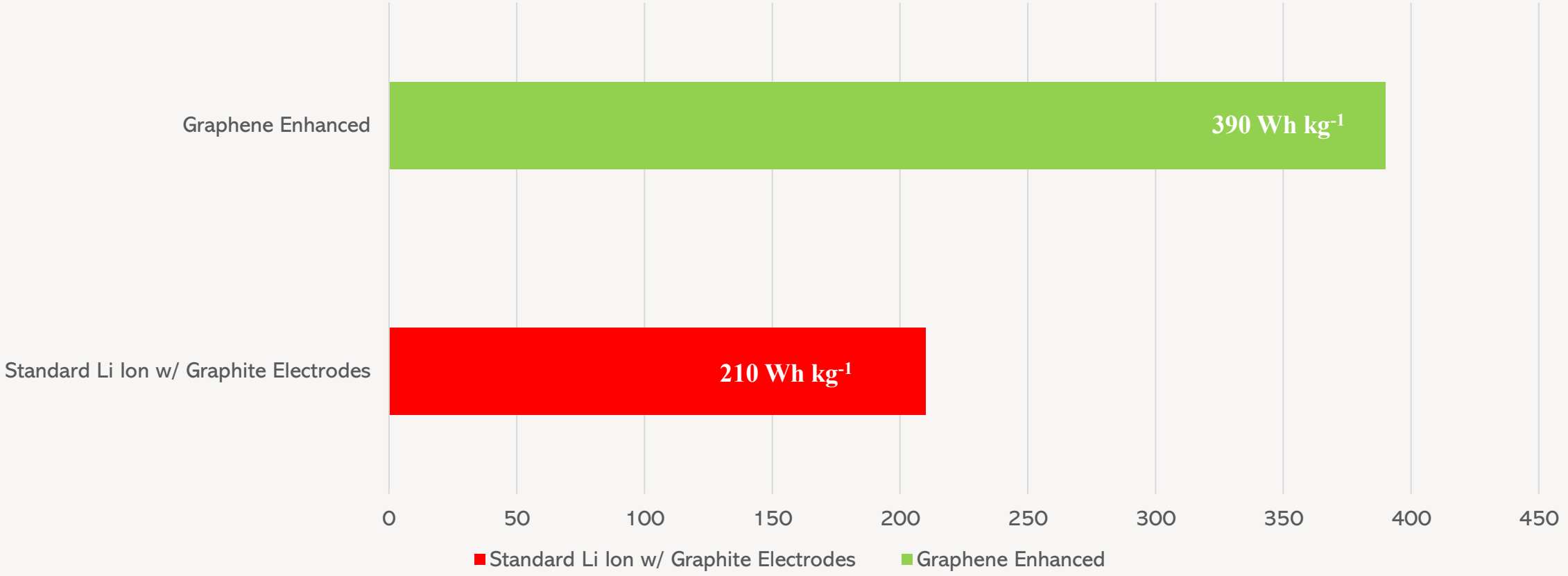
Li-Ion Anode Capacity

Anode capacity means the maximum charge an anode material can store and release per gram.



Energy Density

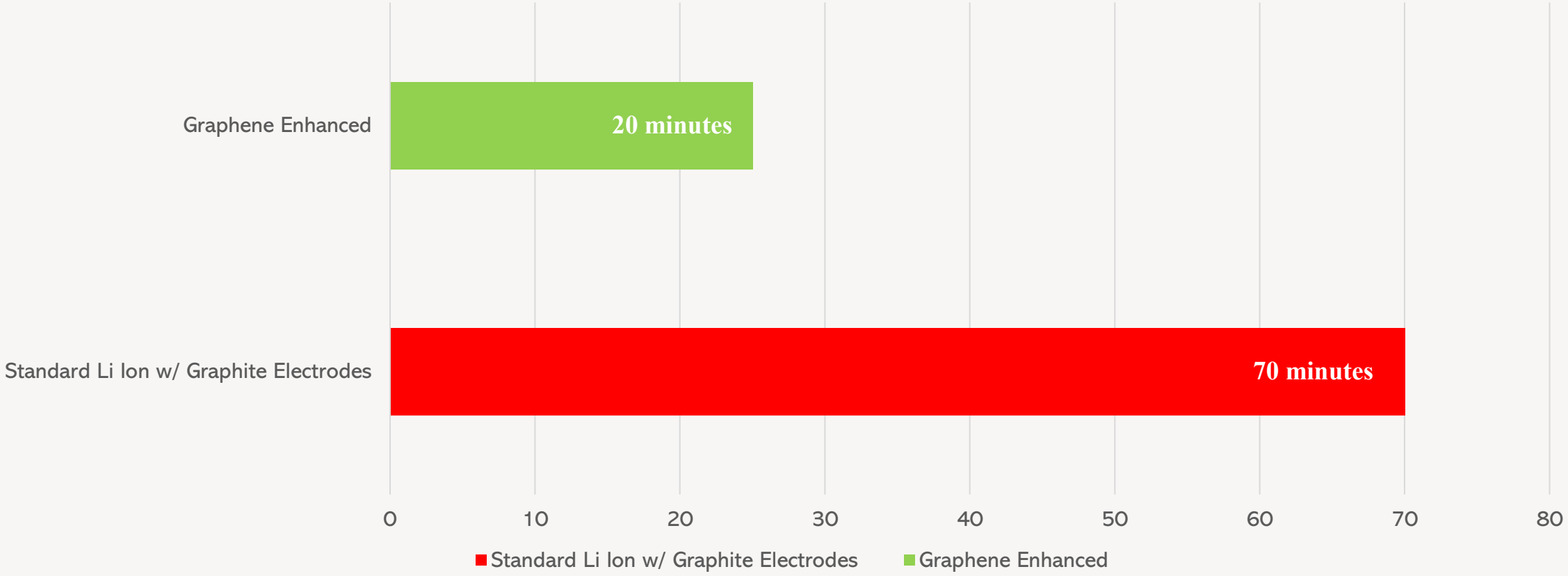
Energy density is the amount of energy that a device can store relative to its weight.



approximate values

Charging Speed

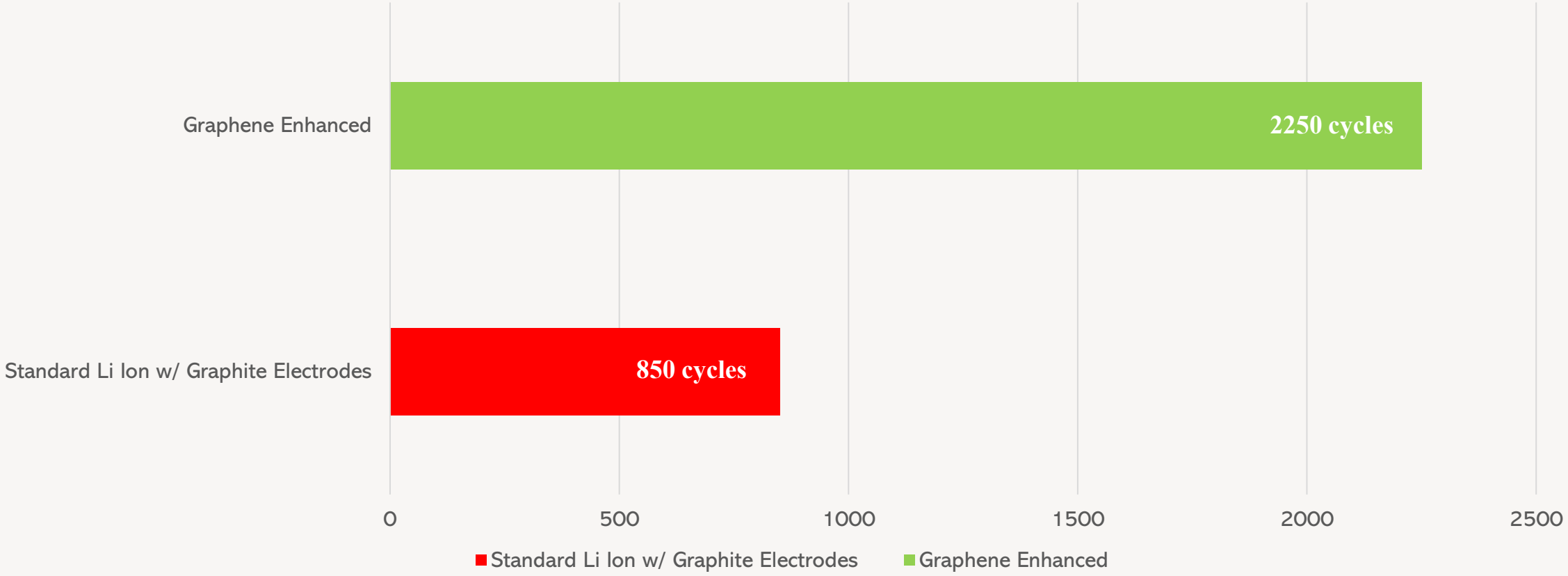
Total time taken to replenish a discharged or partially discharge battery



average values for a 5000 mAh unit

Cycle Life

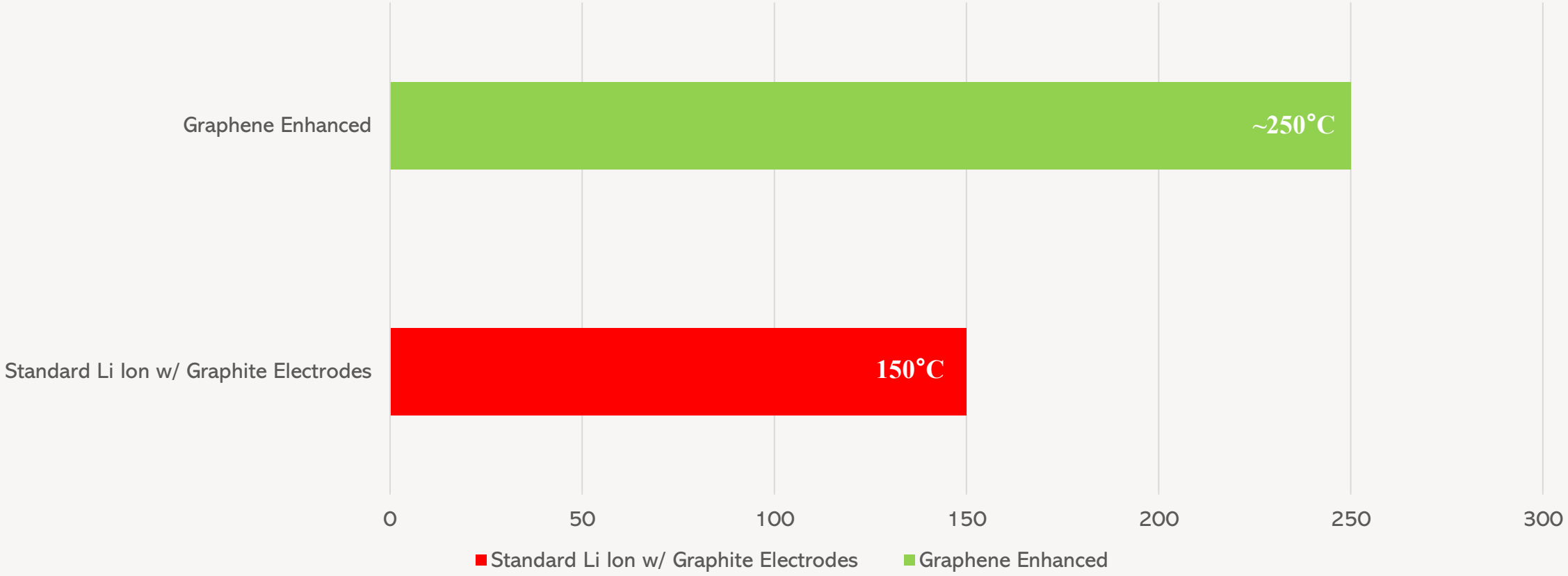
No. of full charge/discharge cycles that a rechargeable battery can endure before significantly degrading.



average values for a 5000 mAh unit

Thermal Runaway Limit

Temperature limit for catastrophic chain reaction that can cause temperature spikes, fire and explosions.



average values for a 5000 mAh unit

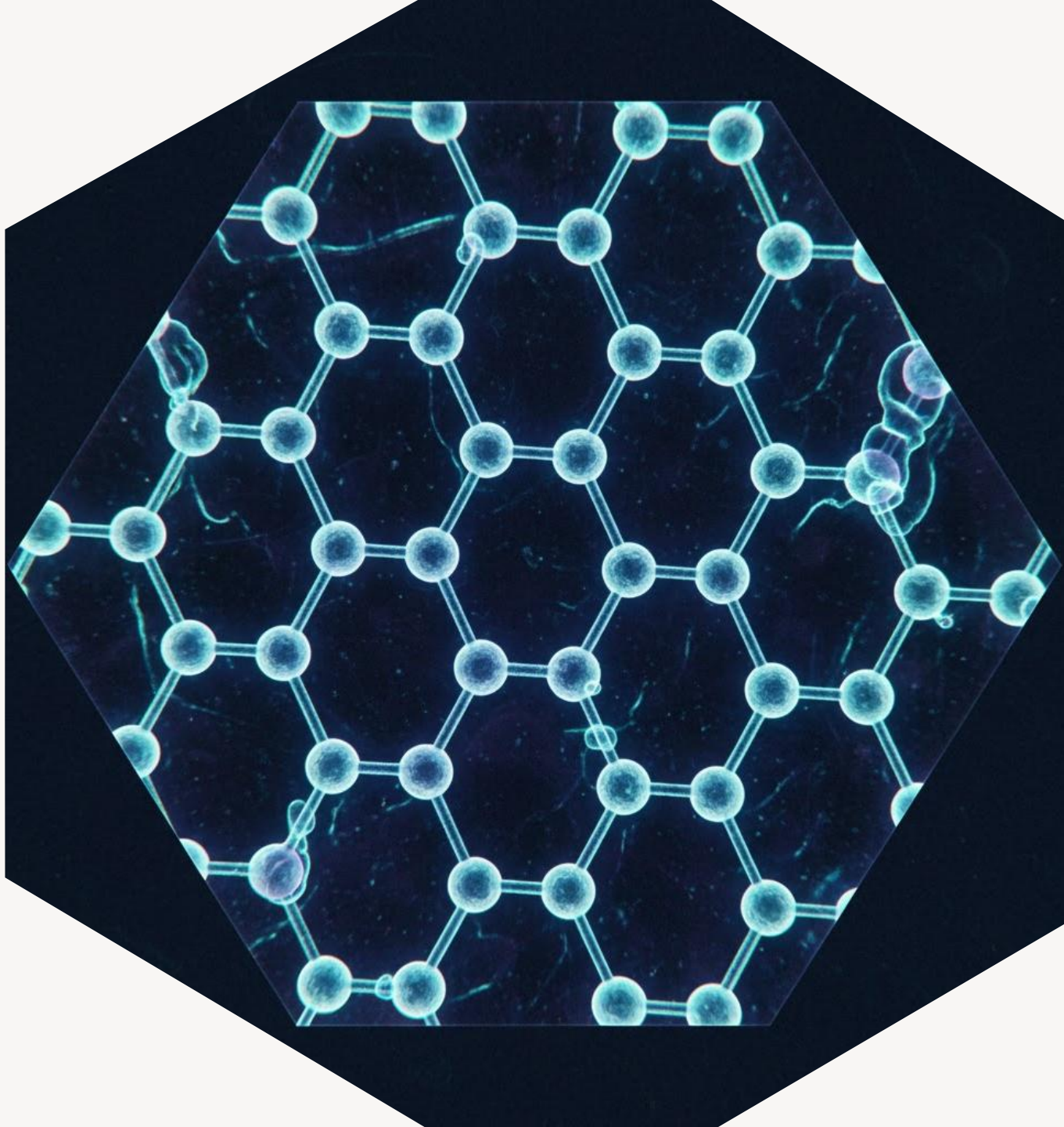
Commercial Devices

1. Silicon-Carbon batteries are emerging as the new standard of energy storage in smartphones.
2. These use graphene as a key-component.
3. These new batteries have almost 50% more capacity than traditional lithium-ion solutions.

Conclusion

The future of energy storage is defined by speed, capacity, and longevity.

Graphene, with its atomic precision and unparalleled properties, is the 2D material that delivers on all three, making it the foundation for our high-performance, sustainable energy future.



THANK YOU